

Topic 11

Chemical Equilibrium

Chemistry in Context

Formation of stalactites — the result of a reversible reaction

Limestone caves are mainly composed of the mineral calcite. Calcite is made up of calcium carbonate.

When water containing dissolved carbon dioxide seeps through a limestone cave, a reaction occurs in which the mineral calcite dissolves, giving a calcium hydrogencarbonate solution.

$CaCO_3(s) + H_2O(l) + CO_2(aq) \longrightarrow Ca(HCO_3)_2(aq)$

This solution travels through the cracks of the roof of a cave and drips down. Once the solution comes into contact with the air inside the cave, the chemical reaction that created it is reversed and calcite starts to form around the crack.

$Ca(HCO_3)_2(aq) \longrightarrow CaCO_3(s) + H_2O(l) + CO_2(aq)$

As water continues to drip, the length and thickness of the calcite grows, and eventually a stalactite forms on the ceiling.

Do you recall?

1 Calcium carbonate occurs naturally in three main forms. Name the two other forms besides limestone.
Chalk and marble

2 Besides the reversible reaction involved in the formation of stalactites, suggest another example of reversible reaction.
Any example of reversible reaction

3 What is the difference between an irreversible reaction and a reversible reaction?
An irreversible reaction goes in only one direction while a reversible reaction goes in both directions.

In this topic, you are going to find out more about reversible reactions and equilibrium systems.

Unit

39 An introduction to chemical equilibrium

Unit Key Concepts

- Irreversible and reversible reactions
- Characteristics of a system in dynamic equilibrium
- Calculations involving equilibrium constant, K_c

Pre-study

A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

39.1 Irreversible and reversible reactions

Irreversible reactions

Some processes go in only one direction — like baking bread. Once baked in an oven, there is no way to reverse the process and split a loaf of bread into flour, water and yeast.

Chemical reactions that take place in one direction only are known as **irreversible reactions**. Burning methane is an example.

$$CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(l)$$

Methane burns in air to produce carbon dioxide and water. Once burnt in air, it is impossible to turn the products back to methane and oxygen.

Reversible reactions

Many reactions, however, are reversible. A **reversible reaction** is one that can take place in both the forward and backward directions.

Reaction of nitrogen and hydrogen

One example of a reversible reaction is the reaction of nitrogen and hydrogen to form ammonia. Ammonia decomposes to give nitrogen and hydrogen in the reverse direction.

forward reaction

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

backward reaction

irreversible reaction 不可逆反應　　reversible reaction 可逆反應

The sign '$\rightleftharpoons$' represents the reversibility of the reaction. The reaction going from left to right is called the **forward reaction** whilst the reaction going from right to left is called the **backward reaction**.

Esterification

The formation of ethyl ethanoate from ethanoic acid and ethanol is also a reversible reaction. The reaction is quite slow and the reaction mixture is heated under reflux with concentrated sulphuric acid as a catalyst.

$$\underset{\text{ethanoic acid}}{CH_3COOH(l)} + \underset{\text{ethanol}}{CH_3CH_2OH(l)} \rightleftharpoons \underset{\text{ethyl ethanoate}}{CH_3COOCH_2CH_3(l)} + H_2O(l)$$

The reaction does not go to completion no matter how long the reaction mixture is heated under reflux. This is because ethyl ethanoate reacts with water to reform ethanoic acid and ethanol. Ethanoic acid, ethanol, ethyl ethanoate and water are all present in the reaction mixture.

Practice 39.1

Decide whether each of the following processes is reversible or not.

a) Heating blue copper(II) sulphate crystals to give white anhydrous copper(II) sulphate
Reversible (The white anhydrous copper(II) sulphate returns to blue if we add water.)

b) Rusting of iron
Irreversible

c) Adding sodium to water
Irreversible

d) Freezing water to ice
Reversible

forward reaction 正向反應　　backward reaction 逆向反應

39.2 Equilibrium

The general meaning of the term '**equilibrium**' is a state of balance. The two main types of equilibrium encountered in our everyday life are static equilibrium and dynamic equilibrium. Fig. 39.1 shows two children balance each other on a seesaw. At the balance point (i.e. equilibrium position), no movement of the children or the seesaw occurs. This is a **static equilibrium**.

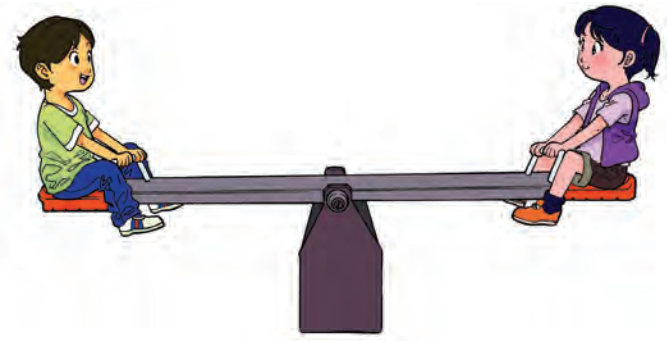


Fig. 39.1 Two children balance each other on a seesaw — a static equilibrium



Fig. 39.2 The girl and the escalator are moving at the same speed in opposite directions; the position of the girl remains unchanged — a dynamic equilibrium

Fig. 39.2 shows a girl moving up the escalator as the escalator is descending. If the girl moves at the same speed as the escalator does, the position of the girl will remain unchanged. This is a **dynamic equilibrium**.

39.3 Chemical equilibrium for a reversible reaction

Take the reversible reaction of hydrogen and iodine at a temperature of 448 °C as an example.

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

Suppose 0.50 mole of hydrogen and 0.50 mole of iodine are sealed in a glass bulb with volume 1.00 dm^3 and kept at a constant temperature of 448 °C. At particular time intervals, the content of the bulb is analyzed.

Fig. 39.3 shows the amount of hydrogen iodide in the bulb at different times. If the reaction went to completion, there would be one mole of hydrogen iodide in the bulb and zero mole of hydrogen and iodine. But this did not happen. Only 0.78 mole of hydrogen iodide is present in the bulb.

equilibrium 平衡　　static equilibrium 靜態平衡　　dynamic equilibrium 動態平衡

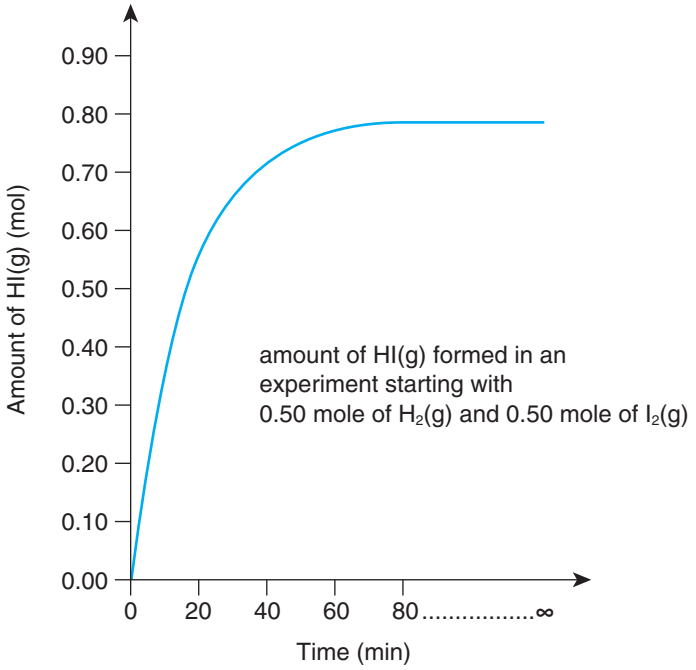


Fig. 39.3 The amount of hydrogen iodide in the bulb at different times

Fig. 39.4 is a different way to present the results. The graph shows the concentration of hydrogen or iodine in red, and the concentration of hydrogen iodide in blue.

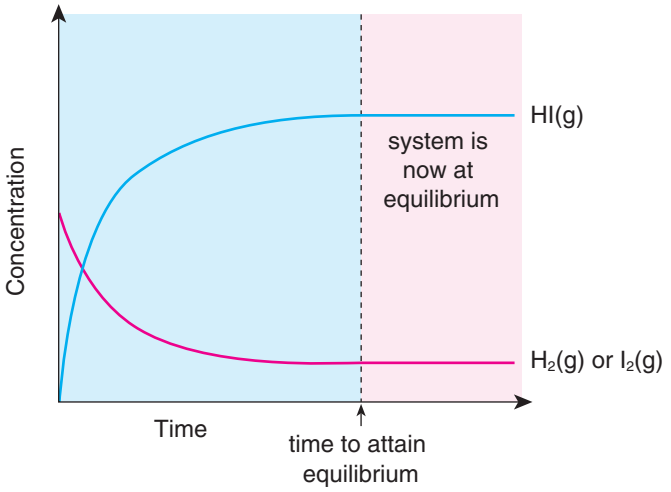


Fig. 39.4 Changes in concentration of reactants and product over time for the reversible reaction between hydrogen and iodine

Reference Website

Visit the following website to run the reversible reaction of hydrogen and iodine ($H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$) and examine the concentration-time plots:
http://www.chm.davidson.edu/vce/Equilibria/BasicConcepts.html
(accessed April 2015)

Dynamic means moving or changing. The word 'dynamic' reminds you that the reaction has not stopped.

The concentrations of the gases do not change once equilibrium has been reached. When equilibrium has been attained, it does not mean the reactions have stopped. It actually means that the forward reaction is making products at the same rate that the backward reaction is making reactants. The reactants and the products are in a state of dynamic equilibrium.

At the start, the rate of the forward reaction is the highest, since the concentrations of the reactants (hydrogen and iodine) are at the highest. As the reaction proceeds, the forward reaction slows down because the reactant concentrations decrease. On the other hand, the rate of the backward reaction increases as the concentration of hydrogen iodide increases, until both rates become equal (Fig. 39.5).

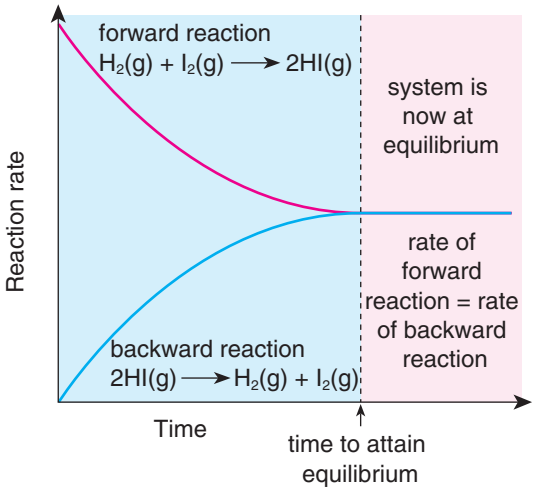


Fig. 39.5 Rate of forward reaction equals rate of backward reaction when the system is at equilibrium

✓ A dynamic equilibrium is attained when the forward and backward reactions occur at the same rate.

Table 39.1 summarizes the rates of the forward and backward reactions during the course of the experiment.

Table 39.1

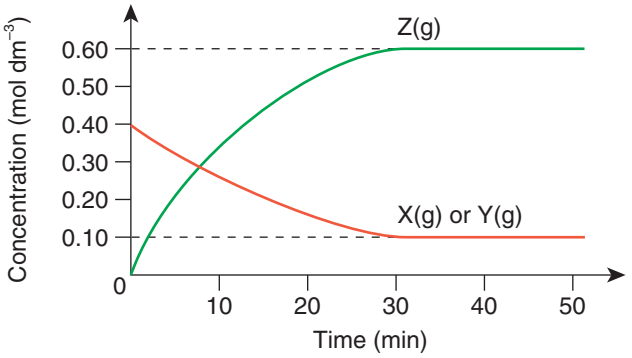
Rates of forward and backward reactions during the experiment

	Rate of forward reaction $H_2(g) + I_2(g) \longrightarrow 2HI(g)$	**Rate of backward reaction** $2HI(g) \longrightarrow H_2(g) + I_2(g)$
At start	high	zero
As reaction proceeds (before equilibrium)	decreasing	increasing
	but rate of forward reaction > rate of backward reaction	
At equilibrium	rate of forward reaction = rate of backward reaction	

Both the reactants and the products are present when a system is at equilibrium. A mixture of reactants and products in an equilibrium state is called an **equilibrium mixture**. Although the rates of the forward and backward reactions are the same, it does not mean that the concentrations of the reactants and the products are the same at equilibrium.

Practice 39.2

X(g) reacts with Y(g) reversibly to give Z(g). A mixture of X(g) and Y(g) is allowed to react in a 1 dm^3 closed container at a constant temperature. The graph below shows the changes in concentrations of X(g), Y(g) and Z(g) in the container with time.



a) From the curves, 1 mole of X(g) reacts with 1 mole of Y(g) to give 2 moles of Z(g).
i.e. $X(g) + Y(g) \rightleftharpoons 2Z(g)$

a) With reference to the above graph, deduce the chemical equation for the reaction.

b) Compare the rate of the forward reaction and that of the backward reaction

i) at the 5th minute and 10th minute after X(g) and Y(g) are mixed;
At the 5th minute and 10th minute, forward reaction rate is greater than backward reaction rate.

ii) at the 40th minute after X(g) and Y(g) are mixed.
At the 40th minute, forward reaction rate is equal to backward reaction rate.

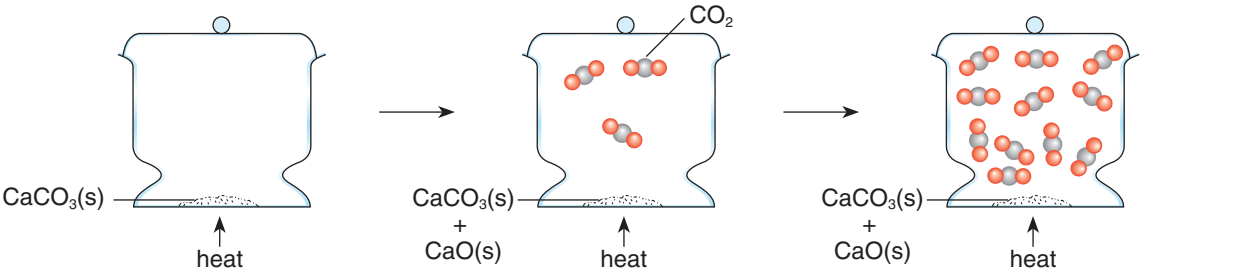
39.4 The importance of a closed system

The glass bulb in the reaction between hydrogen and iodine in the above experiment is sealed so that none of the gases in the reaction can escape. This is an example of a **closed system**. A closed system is one in which no substances can enter or leave the system.

equilibrium mixture 平衡混合物　　closed system 密閉體系

Consider heating calcium carbonate in a closed vessel (Fig. 39.6). The reversible reaction is represented by the following chemical equation and the system will attain an equilibrium state after a period of time.

$$CaCO_3(s) \overset{\text{heat}}{\rightleftharpoons} CaO(s) + CO_2(g)$$



(a) start heating calcium carbonate

(b) calcium oxide and carbon dioxide forming

(c) no more concentration change occurs, equilibrium is attained

Fig. 39.6 Heating calcium carbonate in a closed vessel (the size of carbon dioxide molecule is exaggerated)

If we take the lid off, the system changes into an open system and all the carbon dioxide escapes. As a result, the backward reaction cannot occur as one of the products in the forward reaction is gone. Equilibrium can no longer be attained in this situation and all the calcium carbonate decomposes to form the products.

39.5 Equilibrium attained from either direction of a reaction

Consider the reversible reaction between pale yellow dinitrogen tetroxide gas and dark brown nitrogen dioxide gas:

$$\underset{\text{pale yellow}}{N_2O_4(g)} \rightleftharpoons \underset{\text{dark brown}}{2NO_2(g)}$$

Suppose we seal 0.750 mole of $N_2O_4(g)$ in a 1.00 dm^3 glass bulb and 1.50 moles of $NO_2(g)$ in another bulb of the same volume, both kept at the same temperature. The closed systems allow the reactions to attain equilibrium states.

This is exactly the amount of $NO_2(g)$ that 0.750 mole of $N_2O_4(g)$ would produce if $N_2O_4(g)$ dissociates completely to form $NO_2(g)$.

Table 39.2 lists the initial and equilibrium concentrations of substances in the glass bulbs.

Table 39.2

The initial and equilibrium concentrations of substances in the glass bulbs

	Initial concentration (mol dm^{-3})		Equilibrium concentration (mol dm^{-3})	
	$[N_2O_4(g)]$	$[NO_2(g)]$	$[N_2O_4(g)]$	$[NO_2(g)]$
Bulb 1	0.750	0	0.721	0.0580
Bulb 2	0	1.50	0.721	0.0580

Square brackets are used as symbols to show concentration values measured in mol dm^{-3}.

The concentrations of gases at equilibrium are the same regardless whether the system starts with dinitrogen tetroxide gas or nitrogen dioxide gas (Fig. 39.7).

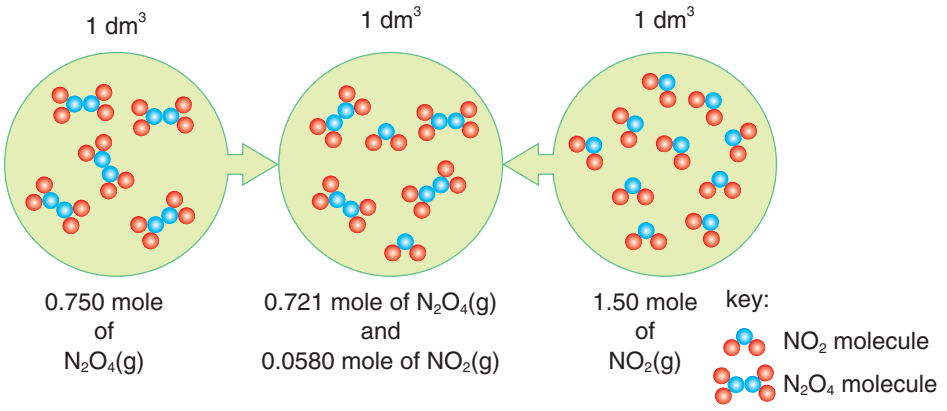


Fig. 39.7 Initial and equilibrium concentrations of systems in the two glass bulbs (This is only a schematic sketch. It does not show the actual numbers and sizes of the molecules.)

From the above experiment, we can see that equilibrium can be attained from either direction, either starting with the substance on the left of the equation (i.e. $N_2O_4(g)$) or with the one on the right ($NO_2(g)$).

39.6 Effect of changing conditions on chemical equilibrium systems

39.1

Investigating the effects of concentration changes on two chemical equilibrium systems.

Suppose a system has attained equilibrium under certain conditions (temperature, pressure and concentration). If the conditions are changed, the equilibrium may be affected.

Effect of concentration changes on the chromate / dichromate system

Consider the equilibrium system attained in an aqueous solution of potassium chromate.

$$\underset{\text{yellow}}{2CrO_4^{2-}(aq)} + 2H^+(aq) \rightleftharpoons \underset{\text{orange}}{Cr_2O_7^{2-}(aq)} + H_2O(l)$$

When a small amount of dilute sulphuric acid is added to the system, the colour of the solution changes gradually from yellow to orange. This indicates that the concentration of $CrO_4^{2-}(aq)$ ions has decreased while that of $Cr_2O_7^{2-}(aq)$ ions has increased (Fig. 39.8).

When a small amount of dilute sodium hydroxide solution is added to the resulting solution, the yellow colour of the solution is restored. This indicates that the concentration of $CrO_4^{2-}(aq)$ ions has increased while that of $Cr_2O_7^{2-}(aq)$ ions has decreased (Fig. 39.8).

At this stage, it is sufficient for you to realize that a chemical equilibrium system can be affected by changes in conditions. We will discuss how changes in conditions affect a chemical equilibrium system in detail in Unit 40.

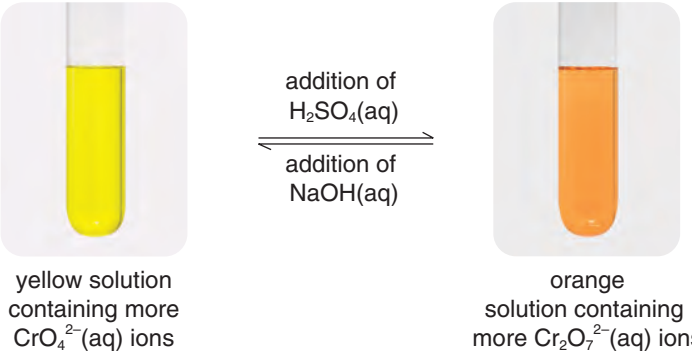


Fig. 39.8 Effect of concentration changes on the chromate / dichromate system

39.7 Characteristics of a system in dynamic equilibrium

Animation
Illustrating dynamic equilibrium

From the above discussion, we can summarize characteristics of a system in dynamic equilibrium as follows:

- Equilibrium can only be established in a closed system with temperature and pressure held constant.
- Once equilibrium is attained, the composition of the system remains constant.
- The reaction has not stopped, rate of forward reaction = rate of backward reaction.
- Reactants and products will both be present in the equilibrium mixture.

- Equilibrium can be attained from either direction, starting with the substances on the left of the equation or starting with the substances on the right.
- If the conditions (temperature, pressure and concentration) are changed, the equilibrium established may be affected.

39.8 The equilibrium constant

Consider again the reversible reaction represented by the following equation:

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

Table 39.3 shows the results of studying the above reaction. In the experiments, different initial amounts of the substances were used and allowed to attain equilibrium. Their concentrations were then measured.

Table 39.3

Data for the reaction involving H_2, I_2 and HI (at 458 °C)
$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

Experiment no.	Initial concentration (mol dm^{-3})			Equilibrium concentration (mol dm^{-3})			$\frac{[HI(g)]^2}{[H_2(g)][I_2(g)]}$
	$[H_2(g)]$	$[I_2(g)]$	$[HI(g)]$	$[H_2(g)]$	$[I_2(g)]$	$[HI(g)]$	
1	2.40×10^{-2}	1.38×10^{-2}	0	1.14×10^{-2}	0.12×10^{-2}	2.52×10^{-2}	46.4
2	2.44×10^{-2}	1.98×10^{-2}	0	0.77×10^{-2}	0.31×10^{-2}	3.34×10^{-2}	46.7
3	0	0	3.04×10^{-2}	0.345×10^{-2}	0.345×10^{-2}	2.35×10^{-2}	46.4
4	0	0	7.58×10^{-2}	0.86×10^{-2}	0.86×10^{-2}	5.86×10^{-2}	46.4

The fraction $\frac{[HI(g)]^2}{[H_2(g)][I_2(g)]}$ in the final column of Table 39.3 gives a reasonably constant value at equilibrium. This value is called the **equilibrium constant** for the reaction in terms of concentration and is given a symbol K_c.

For the above reaction at 458 °C, no matter what initial concentrations of the three substances are used, the equilibrium constant will still be the same.

At a given temperature, the equilibrium constant, K_c, for a specific reaction always has the same value. We must give the temperature when citing a value of K_c.

equilibrium constant 平衡常數

39.9 The equilibrium law

From the above case, we can see that the expression of K_c is derived from the balanced chemical equation. The concentration of the product is placed over the concentrations of the reactants. The coefficients in the chemical equation become the powers in the expression.

For a reversible reaction:

$$aA + bB \rightleftharpoons cC + dD$$

We can calculate the equilibrium constant, K_c, from the equilibrium concentrations of substances by using the following expression:

✓ $$K_c = \frac{[C]^c_{eqm}[D]^d_{eqm}}{[A]^a_{eqm}[B]^b_{eqm}}$$

The subscript 'eqm' on the concentration symbols means that the concentrations are equilibrium concentrations — concentrations of substances when the system has attained equilibrium.

Suppose $N_2O_4(g)$ and $NO_2(g)$ exist in equilibrium in a closed container.

$$N_2O_4(g) \rightleftharpoons 2NO_2(g)$$

The equilibrium concentrations are as follows:

$[N_2O_4(g)] = 0.200 \text{ mol dm}^{-3}$ $[NO_2(g)] = 1.60 \text{ mol dm}^{-3}$

We can calculate the equilibrium constant, K_c, from the equilibrium concentrations of substances by using the following expression:

$$K_c = \frac{[NO_2(g)]^2}{[N_2O_4(g)]}$$
$$= \frac{(1.60 \text{ mol dm}^{-3})^2}{0.200 \text{ mol dm}^{-3}}$$
$$= 12.8 \text{ mol dm}^{-3}$$

Reference Website
Refer to the following website for an animation simulating the gaseous reaction 2A ⟶ B in a closed container:
http://www.chm.davidson.edu/ronutt/che115/equkin/equkin.htm
(accessed April 2015)

Units of equilibrium constant

The units of an equilibrium constant, K_c, depend on the powers of concentrations of substances in the expression for the equilibrium constant.

Consider the reaction between hydrogen and iodine:

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

$$K_c = \frac{[HI(g)]^2}{[H_2(g)][I_2(g)]}$$

The units of K_c are given by $\frac{(\text{mol dm}^{-3})^2}{(\text{mol dm}^{-3})(\text{mol dm}^{-3})}$, i.e. no units

K_c has no units in reactions with equal number of moles of particles on both sides of the equation. This is because the concentration units cancel out in the expression.

For reactions with equations in which the numbers of moles of reactants and products are not equal, K_c will have units.

Consider the reaction between nitrogen and hydrogen:

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

$$K_c = \frac{[NH_3(g)]^2}{[N_2(g)][H_2(g)]^3}$$

The units of K_c are given by $\frac{(\text{mol dm}^{-3})^2}{(\text{mol dm}^{-3})(\text{mol dm}^{-3})^3}$,

i.e. $\text{dm}^6\ \text{mol}^{-2}$

Notice that dm^6 is placed first when giving the units. This follows the convention whereby a unit raised to a positive power is placed first.

Relationship of equilibrium constant to the chemical equation N1

N1 Refer to page T11.

There are two common ways of writing the reaction of sulphur dioxide with oxygen. As a result, there are two different expressions for equilibrium constant (K_c), which give different values.

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g) \qquad K_c = \frac{[SO_3(g)]^2}{[SO_2(g)]^2[O_2(g)]}$$

$$SO_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons SO_3(g) \qquad K_c' = \frac{[SO_3(g)]}{[SO_2(g)][O_2(g)]^{\frac{1}{2}}}$$

Notice that $K_c = (K_c')^2$

Thus, in quoting a value for K_c, we must specify the chemical equation on which the expression for K_c is based.

Reversing the equation also changes the expression for K_c.

$2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$ $\qquad K_c'' = \dfrac{[SO_2(g)]^2[O_2(g)]}{[SO_3(g)]^2}$

Notice that $K_c'' = \dfrac{1}{K_c}$

Practice 39.3

For each of the following equilibria, write down the expression for K_c and state the units.

a) $N_2O_4(g) \rightleftharpoons 2NO_2(g)$ $K_c = \dfrac{[NO_2(g)]^2}{[N_2O_4(g)]}$ units: $\dfrac{(\text{mol dm}^{-3})^2}{\text{mol dm}^{-3}} = \text{mol dm}^{-3}$

b) $CH_3COOH(l) + CH_3CH_2OH(l) \rightleftharpoons CH_3COOCH_2CH_3(l) + H_2O(l)$

c) $Ag^+(aq) + 2NH_3(aq) \rightleftharpoons Ag(NH_3)_2^+(aq)$ $K_c = \dfrac{[Ag(NH_3)_2^+(aq)]}{[Ag^+(aq)][NH_3(aq)]^2}$ units: $\dfrac{(\text{mol dm}^{-3})}{(\text{mol dm}^{-3})(\text{mol dm}^{-3})^2} = \text{dm}^6\ \text{mol}^{-2}$

Practice 39.3

b) $K_c = \dfrac{[CH_2COOCH_2CH_3(g)][H_2O(l)]}{[CH_3COOH(l)][CH_3CH_2OH(l)]}$ units: $\dfrac{(\text{mol dm}^{-3})(\text{mol dm}^{-3})}{(\text{mol dm}^{-3})(\text{mol dm}^{-3})} = \text{no unit}$

39.10 Calculating equilibrium constants

Fig 39.9 summarizes the steps we use to calculate the equilibrium constant, K_c, for a reaction.

1. Write the balanced equation for the reaction.
2. Calculate the quantity of each species at equilibrium using the balanced equation.
3. Calculate the equilibrium concentration of each species.
4. Substitute the equilibrium concentrations into the expression for K_c and calculate K_c.

Fig. 39.9 Steps used to calculate the equilibrium constant, K_c, for a reaction

Example 39.1

Q Ethanol is manufactured by the catalytic hydration of ethene:

$$C_2H_4(g) + H_2O(g) \rightleftharpoons C_2H_5OH(g)$$

In a simulation study of the manufacturing process, 1.64 moles of $C_2H_4(g)$ and 1.00 mole of $H_2O(g)$ were mixed in a 2.00 dm^3 container to react in the presence of a catalyst at 573 K and 60 atmospheres. When equilibrium was attained, the concentration of $C_2H_5OH(g)$ was 0.0410 mol dm^{-3}.

Calculate the equilibrium constant, K_c, for the reaction.

A Number of moles of C_2H_5OH produced when equilibrium was attained
= 0.0410 mol dm^{-3} x 2.00 dm^3
= 0.0820 mol

According to the equation, 1 mole of C_2H_4 reacts with 1 mole of H_2O to produce 1 mole of C_2H_5OH.

i.e. number of moles of C_2H_4 or H_2O reacted = 0.0820 mol

	$C_2H_4(g)$	+	$H_2O(g)$	$\rightleftharpoons$	$C_2H_5OH(g)$
Equilibrium concentration (mol dm^{-3})	$\frac{1.64 - 0.0820}{2.00}$ = 0.779		$\frac{1.00 - 0.0820}{2.00}$ = 0.459		0.0410

$$K_c = \frac{[C_2H_5OH(g)]}{[C_2H_4(g)][H_2O(g)]}$$
$$= \frac{0.0410 \text{ mol dm}^{-3}}{(0.779 \text{ mol dm}^{-3})(0.459 \text{ mol dm}^{-3})}$$
$$= 0.115 \text{ dm}^3 \text{ mol}^{-1}$$

$\therefore$ the equilibrium constant, K_c, for the reaction is 0.115 dm^3 mol^{-1}.

Example 39.2

DSE 2014 Paper 1B Q13(a)(i)

Q A sample of hydrogen iodide (9.00×10^{-3} mol) was placed in an empty 3.00 dm^3 closed container at 800 °C. After equilibrium was reached, 26.0% of the hydrogen iodide had dissociated to give hydrogen and iodine.

$$2HI(g) \rightleftharpoons H_2(g) + I_2(g)$$

Calculate the equilibrium constant, K_c, at 800 °C for the reaction. N2

N2 Examination questions often give the percentage dissociation of a substance and ask students to calculate the equilibrium constant.

Continued on next page

A1 Number of moles of HI dissociated = 9.00×10^{-3} mol x 26.0%
$= 2.34 \times 10^{-3}$ mol

According to the equation, 2 moles of HI decompose to give 1 mole of H_2 and 1 mole of I_2.

i.e. number of moles of H_2 or I_2 formed = $\frac{2.34 \times 10^{-3}}{2}$ mol
$= 1.17 \times 10^{-3}$ mol

	$2HI(g)$	$\rightleftharpoons$	$H_2(g)$	+	$I_2(g)$
Equilibrium concentration (mol dm^{-3})	$\frac{9.00 \times 10^{-3} - 2.34 \times 10^{-3}}{3.00}$ $= 2.22 \times 10^{-3}$		$\frac{1.17 \times 10^{-3}}{3.00}$ $= 3.90 \times 10^{-4}$		$\frac{1.17 \times 10^{-3}}{3.00}$ $= 3.90 \times 10^{-4}$

$$K_c = \frac{[H_2(g)][I_2(g)]}{[HI(g)]^2} = \frac{(3.90 \times 10^{-4} \text{ mol dm}^{-3})(3.90 \times 10^{-4} \text{ mol dm}^{-3})}{(2.22 \times 10^{-3} \text{ mol dm}^{-3})^2}$$
$$= 3.09 \times 10^{-2}$$

$\therefore$ the equilibrium constant, K_c, at 800 °C for the reaction is 3.09×10^{-2}.

Practice 39.4

Refer to page T12.

1 0.0100 mole of phosphorus pentachloride was placed in a 1.00 dm^3 flask at 523 K. It partially dissociated according to the equation:

$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

At equilibrium, the concentration of chlorine was 0.00830 mol dm^{-3}.

a) What were the equilibrium concentrations of $PCl_5(g)$ and $PCl_3(g)$? 0.00170 mol dm^{-3}; 0.00830 mol dm^{-3}

b) Calculate the equilibrium constant, K_c, at 523 K for the reaction. 0.0405 mol dm^{-3}

2 Consider the reaction represented by the equation below:

$Fe^{3+}(aq)$ + $SCN^-(aq)$ $\rightleftharpoons$ $Fe(SCN)^{2+}(aq)$
yellow-brown colourless deep red

In an experiment, 50.0 cm^3 of 0.0160 mol dm^{-3} $Fe_2(SO_4)_3(aq)$ and 50.0 cm^3 of 0.0160 mol dm^{-3} KSCN(aq) were mixed at room temperature, and equilibrium was attained.

The concentration of $Fe(SCN)^{2+}(aq)$ ions in the equilibrium mixture was 0.00724 mol dm^{-3}.

a) What were the initial concentrations of $Fe^{3+}(aq)$ and $SCN^-(aq)$ ions after mixing? 0.0160 mol dm^{-3}; 0.00800 mol dm^{-3}

b) What were the equilibrium concentrations of $Fe^{3+}(aq)$ and $SCN^-(aq)$ ions? 0.00876 mol dm^{-3}; 0.00076 mol dm^{-3}

c) Calculate the equilibrium constant, K_c, at room temperature for the reaction. 1 090 dm^3 mol^{-1}

DSE 2012 Paper 1B Q13(a)

39.11 What does the equilibrium constant tell us?

Knowing the equilibrium constant, K_c, for a chemical reaction lets us judge the extent of the reaction, predict the direction of the reaction and calculate equilibrium concentrations. Let us look at each in turn.

Judging the extent of a reaction

The equilibrium constant, K_c, for a reaction indicates the extent to which reactants are converted to products.

For example, consider the reaction of hydrogen and oxygen, which has a very large equilibrium constant (K_c = 9.1 x 10^{80} dm^3 mol^{-1} at 25 °C):

$$2H_2(g) + O_2(g) \rightleftharpoons 2H_2O(g)$$

$$K_c = \frac{[H_2O(g)]^2}{[H_2(g)]^2[O_2(g)]}$$
$$= 9.1 \times 10^{80}\ dm^3\ mol^{-1}$$

Products appear in the numerator of the expression for K_c and reactants are in the denominator, a very large value of K_c means that the ratio of products to reactants is very large at equilibrium. In other words, the reaction proceeds nearly to completion.

In contrast, if a reaction has a very small value of K_c, the ratio of products to reactants is very small at equilibrium. The reaction hardly proceeds at all when equilibrium is attained.

Table 39.4 shows how the K_c for a reaction can indicate the extent of the reaction.

Table 39.4

K_c for a reaction can indicate the extent of the reaction

Example	Value of K_c	Extent of reaction
$2NO(g) + 2CO(g) \rightleftharpoons N_2(g) + 2CO_2(g)$	~ 10^{59} dm^3 mol^{-1} (at 297 °C)	Reaction goes virtually to completion
$2HCl(g) \rightleftharpoons H_2(g) + Cl_2(g)$	~ 10^{-17} (at 27 °C)	Reaction hardly proceeds at all

Notice that the equilibrium constant tells us nothing about the rate of a reaction. Even if the equilibrium constant for a reaction is very large, the rate at which it happens may be extremely slow. The reaction of hydrogen and oxygen to form water is a good example of this. Even though K_c for the reaction is around 9×10^{80}, a mixture of hydrogen and oxygen at room temperature does not spontaneously react to form water.

Predicting the direction of a reaction

Consider the reaction of hydrogen with iodine again:

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g) \qquad K_c = 46.4 \text{ at } 458\ ^\circ C$$

Suppose we have a mixture of hydrogen, iodine and hydrogen iodide at 458 °C with the following concentrations:

$[H_2(g)]_t = 0.100 \text{ mol dm}^{-3}$

$[I_2(g)]_t = 0.100 \text{ mol dm}^{-3}$

$[HI(g)]_t = 0.400 \text{ mol dm}^{-3}$

The subscript 't' on the concentration symbols means that the concentrations are measured at some arbitrary time *t*, not necessarily at equilibrium.

If we substitute these concentrations into the expression for equilibrium constant, K_c, we obtain a value called the **reaction quotient**, Q_c.

$$\text{Reaction quotient, } Q_c = \frac{[HI(g)]_t^2}{[H_2(g)]_t[I_2(g)]_t}$$

$$= \frac{(0.400 \text{ mol dm}^{-3})^2}{(0.100 \text{ mol dm}^{-3})(0.100 \text{ mol dm}^{-3})}$$

$$= 16.0$$

The expression for the reaction quotient, Q_c, is defined in the same way as the equilibrium constant except that the concentrations of reactants and products can be taken at any moment of the reaction (not necessarily the concentrations at equilibrium). Q_c is not a constant.

For the case at hand, Q_c (16.0) does not equal to K_c (46.4), so the mixture is not at equilibrium. Hydrogen and iodine would react to form hydrogen iodide so that $[H_2(g)]$ and $[I_2(g)]$ decrease while [HI(g)] increases. This causes Q_c to increase. Equilibrium is attained when $Q_c = K_c$ (Fig. 39.10).

reaction quotient 反應商數

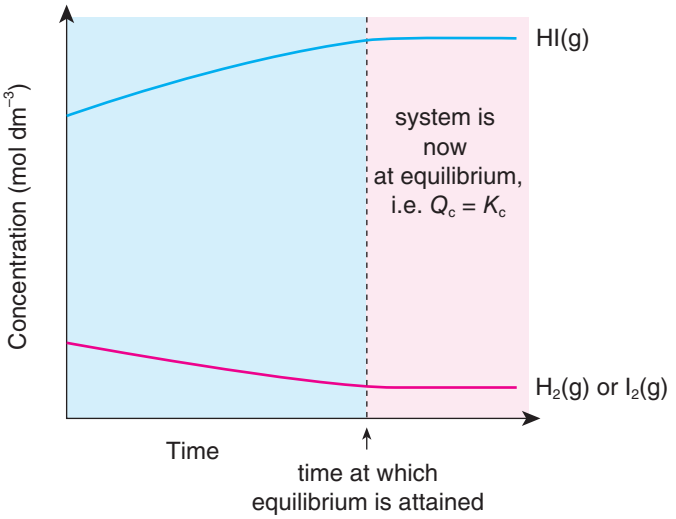


Fig. 39.10 As $Q_c < K_c$, $[H_2(g)]$ and $[I_2(g)]$ must decrease while [HI(g)] must increase until $Q_c = K_c$

We can predict the direction in which the net reaction will proceed to attain equilibrium by comparing Q_c and K_c. There are three possible cases:

- If $Q_c = K_c$, then the system is at equilibrium (Fig. 39.11a).
- If $Q_c < K_c$, then the system is not at equilibrium. Concentrations of the products must increase and that of the reactants must decrease until $Q_c = K_c$. Thus, a net forward reaction◂ occurs until equilibrium is attained (Fig. 39.11b).
- If $Q_c > K_c$, then the system is also not at equilibrium. Concentrations of the products must decrease and that of the reactants must increase until $Q_c = K_c$. Thus, a net backward reaction◂ occurs until equilibrium is attained (Fig. 39.11c).

◂ In this case, both the forward and backward reaction occur; but the rate of forward reaction > the rate of backward reaction.

◂ In this case, both the forward and backward reaction occur; but the rate of backward reaction > the rate of forward reaction.

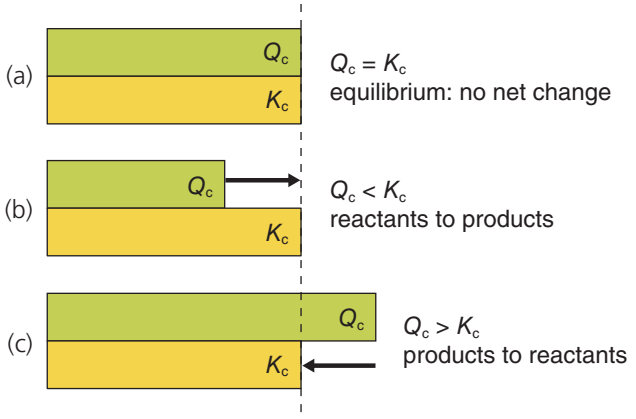


Fig. 39.11 The direction of a reaction can be predicted by comparing Q_c and K_c

Calculating equilibrium concentrations

If the equilibrium constant, K_c, and all the equilibrium concentrations but one are known, we can calculate the unknown concentration by using the expression for K_c.

Example 39.3

Q At 30 °C, the equilibrium constant for the decomposition of phosphorus pentachloride into phosphours trichloride and chlorine is 4.48×10^{-2} mol dm^{-3}.

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$$

A sample of phosphorus pentachloride was placed in a container kept at 30 °C. When equilibrium was attained, the concentration of phosphorus trichloride was 0.150 mol dm^{-3}. Calculate the equilibrium concentration of phosphorus pentachloride.

A Let x mol dm^{-3} be the equilibrium concentration of $PCl_5(g)$.

According to the equation, 1 mole of PCl_5 decomposes to give 1 mole of PCl_3 and 1 mole of Cl_2.

i.e. $[PCl_3(g)]_{eqm} = [Cl_2(g)]_{eqm} = 0.150$ mol dm^{-3}

$$K_c = 4.48 \times 10^{-2} \text{ mol dm}^{-3} = \frac{[PCl_3(g)][Cl_2(g)]}{[PCl_5(g)]}$$

$$= \frac{(0.150 \text{ mol dm}^{-3})(0.150 \text{ mol dm}^{-3})}{x \text{ mol dm}^{-3}}$$

$$x = 0.502$$

$\therefore$ the equilibrium concentration of phosphorus pentachloride is 0.502 mol dm^{-3}.

Example 39.4

ASL 2008 Paper 2 Q4(a)(i)

Q At 25 °C, the equilibrium constant, K_c, for the following reaction is 5.00.

$$CH_3COOH(l) + CH_3CH_2OH(l) \rightleftharpoons CH_3COOCH_2CH_3(l) + H_2O(l)$$

A mixture of 0.350 mole of ethanoic acid and 0.350 mole of ethanol and a few drops of concentrated sulphuric acid was allowed to attain equilibrium at 25 °C.

Calculate the number of moles of ethyl ethanoate in the equilibrium mixture.

A Suppose there was n mole of ethyl ethanoate in the equilibrium mixture. Let V dm^3 be the total volume of the reaction mixture.

	$CH_3COOH(l)$	+	$CH_3CH_2OH(l)$	$\rightleftharpoons$	$CH_3COOCH_2CH_3(l)$	+	$H_2O(l)$
Equilibrium concentration (mol dm^{-3})	$\frac{0.350 - n}{V}$		$\frac{0.350 - n}{V}$		$\frac{n}{V}$		$\frac{n}{V}$

Continued on next page

$$K_c = 5.00 = \frac{[CH_3COOCH_2CH_3(l)][H_2O(l)]}{[CH_3COOH(l)][CH_3CH_2OH(l)]}$$

$$= \frac{\left(\frac{n}{V}\ \text{mol dm}^{-3}\right)\left(\frac{n}{V}\ \text{mol dm}^{-3}\right)}{\left(\frac{0.350 - n}{V}\ \text{mol dm}^{-3}\right)\left(\frac{0.350 - n}{V}\ \text{mol dm}^{-3}\right)}$$

$$= \frac{n^2}{(0.350 - n)^2}$$

Rearranging the equation gives

$$\frac{n}{(0.350 - n)} = \sqrt{5.00}$$

$$n = 0.242$$

Notice that we can calculate the answer without knowing the total volume (V dm^3) of the reaction mixture. This is because there are equal numbers of molecules on both sides of the equation and the volume terms in the expression for K_c cancel out.

$\therefore$ there was 0.242 mole of ethyl ethanoate in the equilibrium mixture.

Example 39.5

Q The equilibrium constant, K_c, for the following reaction is 2.50 dm^3 mol^{-1} at 1 000 K.

$$CO(g) + Cl_2(g) \rightleftharpoons COCl_2(g)$$

A mixture of 2.00 moles of CO(g), 1.00 mole of Cl_2(g) and 4.00 moles of $COCl_2$(g) is introduced into an evacuated vessel of 2.00 dm^3 kept at 1 000 K.

a) i) For this system under the initial conditions, calculate its reaction quotient.

ii) Predict and explain, under the initial conditions, whether the forward reaction rate or the backward reaction rate is greater.

b) Calculate the concentration of $COCl_2$(g) when the system attains equilibrium at 1 000 K.

DSE 2013 Paper 1B Q12(a) & (b)

A a) i) $Q_c = \frac{[COCl_2(g)]}{[CO(g)][Cl_2(g)]}$

$$= \frac{\frac{4.00}{2.00}\ \text{mol dm}^{-3}}{\left(\frac{2.00}{2.00}\ \text{mol dm}^{-3}\right)\left(\frac{1.00}{2.00}\ \text{mol dm}^{-3}\right)}$$

$$= 4.00\ \text{dm}^3\ \text{mol}^{-1}$$

ii) As $Q_c > K_c$, $[COCl_2(g)]$ must decrease while $[CO(g)]$ and $[Cl_2(g)]$ must increase until $Q_c = K_c$.

A net backward reaction occurs, i.e. the backward reaction rate is greater than the forward reaction rate.

Continued on next page

b) Let x be the number of moles of $COCl_2(g)$ dissociated when the system attains equilibrium.

	$CO(g)$	+	$Cl_2(g)$	$\rightleftharpoons$	$COCl_2(g)$
Equilibrium concentration (mol dm^{-3})	$\frac{2.00 + x}{2.00}$		$\frac{1.00 + x}{2.00}$		$\frac{4.00 - x}{2.00}$

$$K_c = 2.50 \text{ dm}^3 \text{ mol}^{-1} = \frac{\frac{4.00 - x}{2.00} \text{ mol dm}^{-3}}{\left(\frac{2.00 + x}{2.00} \text{ mol dm}^{-3}\right)\left(\frac{1.00 + x}{2.00} \text{ mol dm}^{-3}\right)}$$

$$2.50 = \frac{2(4.00 - x)}{(2.00 + x)(1.00 + x)}$$

Rearranging the equation gives

$2.50(2.00 + x)(1.00 + x) = 2(4.00 - x)$

$2.50x^2 + 7.50x + 5.00 = 8.00 - 2x$

$2.50x^2 + 9.50x - 3.00 = 0$

The equation is in the standard quadratic form, $ax^2 + bx + c = 0$. We can solve for x using the quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Solving the quadratic equation gives two solutions:

$x = 0.293$ or -4.09 (rejected)

$$\text{Equilibrium concentration of } COCl_2 = \frac{(4.00 - 0.293) \text{ mol}}{2.00 \text{ dm}^3} = 1.85 \text{ mol dm}^{-3}$$

Practice 39.5

Refer to page T13.

1 For the reaction

$2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$

the equilibrium constant at 727 °C is 6.25×10^{-3} mol dm^{-3}.

A sample of $SO_3(g)$ was heated to 727 °C. At equilibrium, the concentration of $SO_3(g)$ was 1.60 mol dm^{-3}. Calculate the equilibrium concentration of $O_2(g)$.

0.159 mol dm^{-3}

Continued on next page

ASL 2007 Paper 2 Q4(a)(ii)
ASL 2008 Paper 2 Q4(a)(i)

ASL 2007 Paper 2 Q4(a)(i), (ii)

2 For the reaction

$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

the equilibrium constant is 159 at 500 K.

A 1.00 dm^3 sealed container, which is maintained at 500 K, initiaily contains 0.240 mole of $H_2(g)$, 0.240 mole of $I_2(g)$ and 0.800 mole of HI(g).

a) i) For this system under the initial conditions, calculate its reaction quotient.
11.1

ii) Predict and explain the direction in which the net reaction will proceed to attain equilibrium.

b) Calculate the concentration of $I_2(g)$ when the system attains equilibrium at 500 K.
0.0880 mol dm^{-3}

39.12 Determining the equilibrium constant for an esterification reaction experimentally

Assessment Form
An assessment form on this activity is available for download from *Jing Kung Chemistry Website*.

39.2
Determining the equilibrium constant, K_c, for an esterification reaction.

Now we are going to see how we can determine the equilibrium constant, K_c, for an esterification reaction — the reaction between ethanoic acid and propan-2-ol.

$$CH_3COOH(l) + CH_3CH(OH)CH_3(l) \rightleftharpoons CH_3COOCH(CH_3)_2(l) + H_2O(l)$$

The experiment consists of four steps as listed below (Table 39.5). Sample titration results are shown as well.

Table 39.5
Steps for determining the equilibrium constant for an esterification reaction

	Procedure	Titration result
Step 1	Mix 0.300 mole of ethanoic acid and 0.300 mole of propan-2-ol in a pear-shaped flask.	
Step 2	Withdraw 1.00 cm^3 of this mixture and add to a conical flask containing 25 cm^3 of distilled water. Titrate the contents of the conical flask against 0.250 mol dm^{-3} sodium hydroxide solution (Fig. 39.12).	30.0 cm^3 of NaOH(aq)
Step 3	Add a few drops of concentrated sulphuric acid to the acid-alcohol mixture in the pear-shaped flask. Titrate 1.00 cm^3 of the resulting mixture against 0.250 mol dm^{-3} sodium hydroxide solution as in *Step 2*.	30.1 cm^3 of NaOH(aq)
Step 4	Heat the mixture in the pear-shaped flask under reflux for 2 hours. After cooling, withdraw 1.00 cm^3 of the mixture and titrate against 0.250 mol dm^{-3} sodium hydroxide solution (Fig. 39.13).	14.0 cm^3 of NaOH(aq)

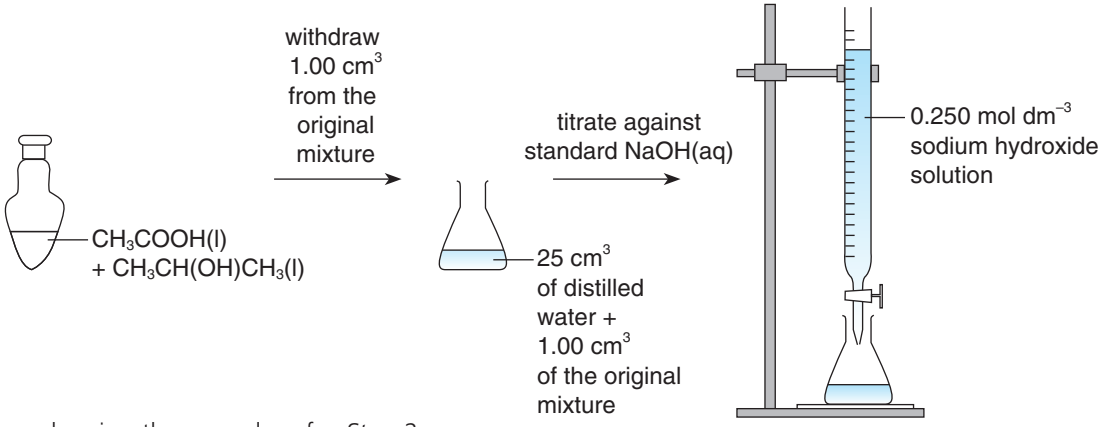


Fig. 39.12 A diagram showing the procedure for *Step 2*

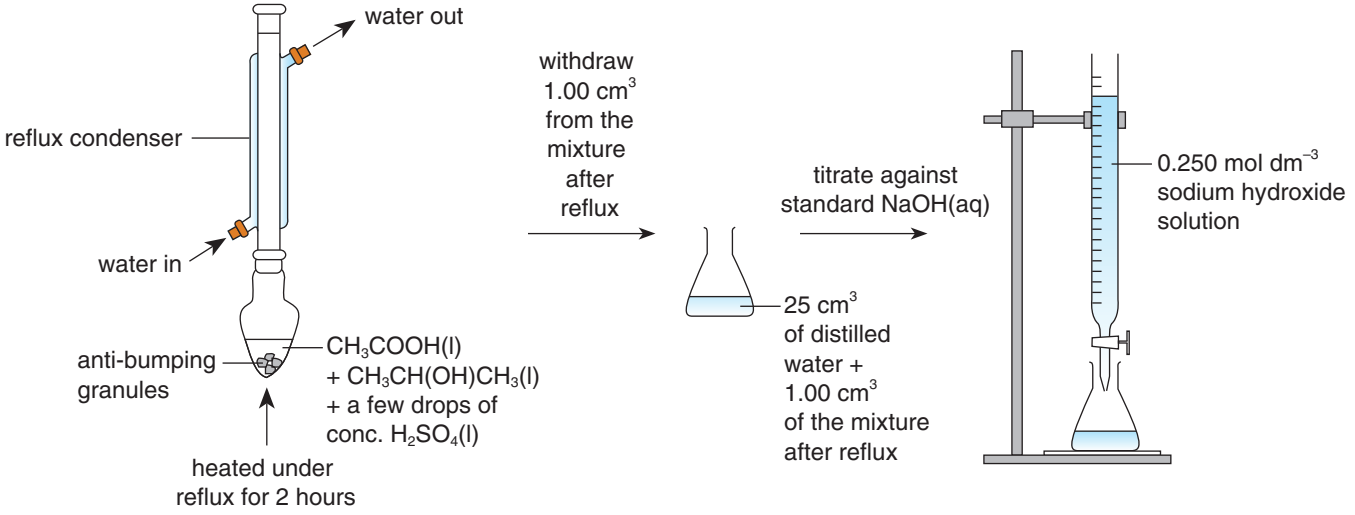


Fig. 39.13 A diagram showing the procedure for *Step 4*

Calculating the equilibrium constant, K_c, for the reaction

Part A — Calculating the concentration of ethanoic acid in the original mixture

30.0 cm^3 of the alkali were required to neutralize the ethanoic acid in 1.00 cm^3 of the original mixture.

NaOH(aq) reacts with $CH_3COOH(l)$ according to the following equation:

$$CH_3COOH(l) + NaOH(aq) \rightleftharpoons CH_3COONa(aq) + H_2O(l)$$

Number of moles of NaOH = number of moles of CH_3COOH

$$0.250 \text{ mol dm}^{-3} \times \frac{30.0}{1\,000} \text{ dm}^3 = [CH_3COOH(l)] \times \frac{1.00}{1\,000} \text{ dm}^3$$

$$[CH_3COOH(l)] = 7.50 \text{ mol dm}^{-3}$$

$\therefore$ concentration of $CH_3COOH(l)$ in the original mixture is 7.50 mol dm^{-3}.

Part B — Calculating the concentration of ethanoic acid in the mixture after reflux

After the reflux, 14.0 cm^3 of the alkali were required to neutralize the ethanoic acid plus the sulphuric acid catalyst in 1.00 cm^3 of the mixture.

From the titration results of *Steps 2* and *3*, we can see that the sulphuric acid catalyst in 1.00 cm^3 of the mixture required 0.1 cm^3 of the alkali for neutralization.

$\therefore$ volume of NaOH(aq) required to neutralize $CH_3COOH(l)$ in 1.00 cm^3 of the mixture after reflux
$= (14.0 - 0.1) \text{ cm}^3 = 13.9 \text{ cm}^3$

Number of moles of NaOH = number of moles of CH_3COOH

$$0.250 \text{ mol dm}^{-3} \times \frac{13.9}{1\,000} \text{ dm}^3 = [CH_3COOH(l)] \times \frac{1.00}{1\,000} \text{ dm}^3$$

$$[CH_3COOH(l)] = 3.48 \text{ mol dm}^{-3}$$

$\therefore$ concentration of $CH_3COOH(l)$ in the mixture after reflux is 3.48 mol dm^{-3}.

Part C — Calculating concentrations of other substances in the mixture after reflux and then the equilibrium constant, K_c

Consider the equation for the esterification reaction:

$$CH_3COOH(l) + CH_3CH(OH)CH_3(l) \rightleftharpoons CH_3COOCH(CH_3)_2(l) + H_2O(l)$$

According to the equation, 1 mole of CH_3COOH reacts with 1 mole of $CH_3CH(OH)CH_3$ to give 1 mole of $CH_3COOCH(CH_3)_2$ and 1 mole of H_2O.

Change in concentration of $CH_3COOH(l)$
= concentration of $CH_3COOH(l)$ in mixture after reflux
– concentration of $CH_3COOH(l)$ in original mixture
= (3.48 – 7.50) mol dm^{-3}
= –4.02 mol dm^{-3}

Notice that a '+' sign denotes an increase in concentration while a '–' sign denotes a decrease in concentration.

Table 39.6 shows concentrations of substances in the mixture after reflux, obtained by using the change in concentration of $CH_3COOH(l)$.

Table 39.6

Concentrations of substances in the mixture after reflux

	Substance			
	$CH_3COOH(l)$	$CH_3CH(OH)CH_3(l)$	$CH_3COOCH(CH_3)_2(l)$	$H_2O(l)$
Concentration in original mixture (mol dm^{-3})	7.50	7.50	0	0
Change in concentration (mol dm^{-3})	–4.02	–4.02	+4.02	+4.02
Concentration in mixture after reflux (mol dm^{-3})	3.48	3.48	4.02	4.02

$$K_c = \frac{[CH_3COOCH(CH_3)_2(l)][H_2O(l)]}{[CH_3COOH(l)][CH_3CH(OH)CH_3(l)]}$$
$$= \frac{(4.02 \text{ mol dm}^{-3})(4.02 \text{ mol dm}^{-3})}{(3.48 \text{ mol dm}^{-3})(3.48 \text{ mol dm}^{-3})}$$
$$= 1.33$$

Notice that K_c has no units because the esterification reaction has equal number of moles of molecules on both sides of the equation.

Notice the following points when determining an equilibrium constant, K_c, experimentally:

- The reactants and products of the reaction must have actually attained equilibrium. This can be tested by taking small samples out of the mixture at different times for concentration analysis. When the same result is obtained from successive analyses, we can assume that there is no further overall change in the mixture, i.e. equilibrium is attained.

- The temperature of the system under study must be known and kept constant.

Have you mastered?

Key terms

irreversible reaction 不可逆反應 2 reversible reaction 可逆反應 2 forward reaction 正向反應 3

backward reaction 逆向反應 3 equilibrium 平衡 4 static equilibrium 靜態平衡 4

dynamic equilibrium 動態平衡 4 equilibrium mixture 平衡混合物 7 closed system 密閉體系 7

equilibrium constant 平衡常數 11 reaction quotient 反應商數 18

Checklist

After studying this unit, you should be able to

- ☐ describe reversible and irreversible reactions by using suitable examples;
- ☐ describe the characteristics of a system in dynamic equilibrium;
- ☐ express the mathematical relationship between concentrations of reactants and products at equilibrium and K_c;
- ☐ perform calculations involving K_c;
- ☐ predict the shift in position of equilibrium using the reaction quotient;
- ☐ perform practical work on the determination of K_c.

(Put a '✔' in the box if you have acquired the knowledge concerned.)

Summary

1 a) Irreversible reactions are chemical reactions that take place in one direction only.

b) Reversible reactions are chemical reactions that take place in both the forward and backward directions.

2 The sign ' $\rightleftharpoons$ ' represents the reversibility of a reaction.

forward reaction

$$aA + bB \rightleftharpoons cC + dD$$

backward reaction

3 A dynamic equilibrium is attained when the forward and backward reactions occur at the same rate.

4 Characteristics of a system in dynamic equilibrium:

- Equilibrium can only be established in a closed system with temperature and pressure held constant.
- Once equilibrium is attained, the composition of the system remains constant.
- The reaction has not stopped, rate of forward reaction = rate of backward reaction.
- Reactants and products will both be present in the equilibrium mixture.
- Equilibrium can be attained from either direction.
- If the conditions (temperature, pressure and concentration) are changed, the equilibrium established may be affected.

Have you mastered?

5 For a reversible reaction:

$$aA + bB \rightleftharpoons cC + dD$$

$$\text{equilibrium constant } (K_c) = \frac{[C]^c_{eqm}[D]^d_{eqm}}{[A]^a_{eqm}[B]^b_{eqm}}$$

6 Steps we can use to calculate the equilibrium constant, K_c, for a reaction:

1 Write the balanced equation for the reaction. → 2 Calculate the quantity of each species at equilibrium using the balanced equation. → 3 Calculate the equilibrium concentration of each species. → 4 Substitute the equilibrium concentrations into the expression for K_c and calculate K_c.

7 For a reversible reaction:

$$aA + bB \rightleftharpoons cC + dD$$

$$\text{reaction quotient } (Q_c) = \frac{[C]_t^c [D]_t^d}{[A]_t^a [B]_t^b}$$

- If $Q_c = K_c$, the system is at equilibrium.
- If $Q_c < K_c$, a net forward reaction occurs until equilibrium is attained.
- If $Q_c > K_c$, a net backward reaction occurs until equilibrium is attained.

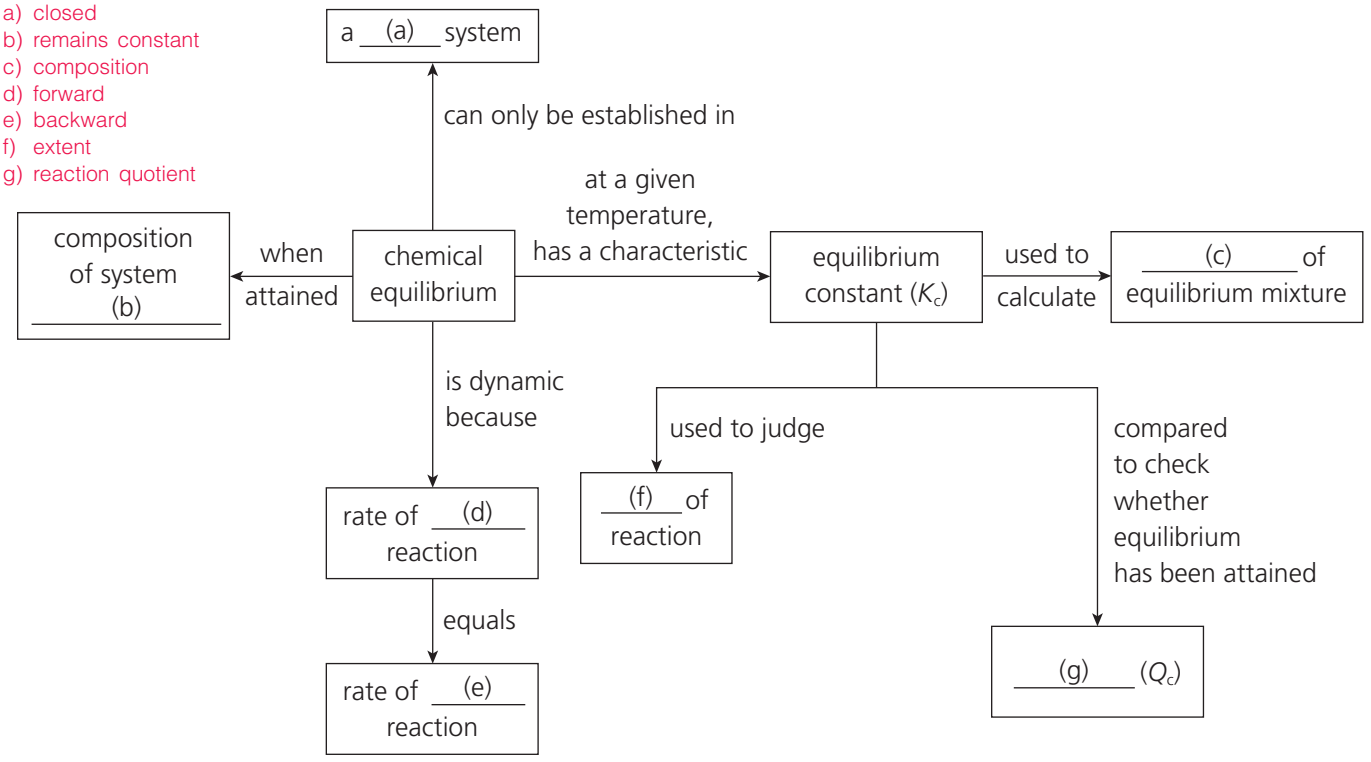
Unit Exercise

Note: The symbol ! indicates the level of difficulty of a question.

Part I Knowledge and understanding

1 Complete the following concept map.

a) closed
b) remains constant
c) composition
d) forward
e) backward
f) extent
g) reaction quotient



2 Write an expression for the equilibrium constant, K_c, for each of the following reactions:

a) $2O_3(g) \rightleftharpoons 3O_2(g)$ $K_c = \dfrac{[O_2(g)]^3}{[O_3(g)]^2}$

b) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ $K_c = \dfrac{[PCl_3(g)][Cl_2(g)]}{[PCl_5(g)]}$

c) $CH_3COOH(l) + C_3H_7OH(l) \rightleftharpoons CH_3COOC_3H_7(l) + H_2O(l)$ $K_c = \dfrac{[CH_3COOC_3H_7(l)][H_2O(l)]}{[CH_3COOH(l)][C_3H_7OH(l)]}$

d) $CH_4(g) + H_2O(g) \rightleftharpoons CO(g) + 3H_2(g)$ $K_c = \dfrac{[CO(g)][H_2(g)]^3}{[CH_4(g)][H_2O(g)]}$

3 a) The following reaction

$PCl_3(g) + Cl_2(g) \rightleftharpoons PCl_5(g)$

has an equilibrium constant, K_c, of about 1 $dm^3\ mol^{-1}$ at 197 °C.

What does this suggest about the extent of the reaction at 197 °C? Comparable concentrations of $PCl_3(g)$, $Cl_2(g)$ and $PCl_5(g)$ are present at equilibrium.

b) The following reaction

$2HCl(g) \rightleftharpoons H_2(g) + Cl_2(g)$

has an equilibrium constant, K_c, of about 10^{-17} at 27 °C.

What does this suggest about the extent of the reaction at 27 °C? The reaction hardly proceeds at all.

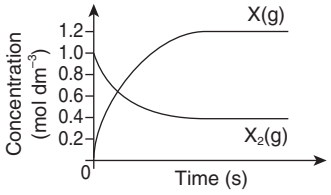
Part II Multiple choice questions

4 In a 1 dm^3 closed container, 1 mole of $X_2(g)$ undergoes decomposition to form X(g) until equilibrium is attained. The chemical equation concerned is shown below:

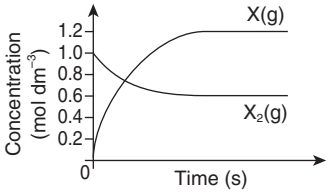
$X_2(g) \rightleftharpoons 2X(g)$

Which of the following graphs correctly shows the variation in concentrations of $X_2(g)$ and X(g) with time?

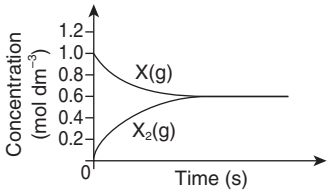
(A)



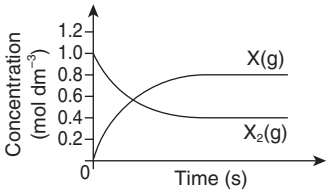
B



C



D



(HKDSE, Paper 1A, 2012, 27)

Directions : Questions 5 and 6 refer to the following information.

The reaction mixture of carbon monoxide and hydrogen attains a dynamic equilibrium.

$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$

5 Which of these statements about a dynamic equilibrium is NOT true?

A The forward rate of reaction is equal to the backward rate of reaction.
B The concentrations of the products and reactants do not change.
(C) The concentrations of the products and reactants are equal.
D The equilibrium can be approached from either direction.

(Edexcel Advanced GCE, Unit 4, Jan. 2012, 1(a))

6 The K_c expression for the above reaction is

(A) $K_c = \dfrac{[CH_3OH(g)]}{[CO(g)] \times [H_2(g)]^2}$

B $K_c = \dfrac{[CO(g)] \times 2[H_2(g)]}{[CH_3OH(g)]}$

C $K_c = \dfrac{[CO(g)] \times [H_2(g)]^2}{[CH_3OH(g)]}$

D $K_c = \dfrac{[CH_3OH(g)]}{[CO(g)] \times 2[H_2(g)]}$

(Edexcel Advanced GCE, Unit 4, Jan. 2012, 1(b))

7 The reaction below reached a dynamic equilibrium from an initial mixture of all four substances, P, Q, R and S, in aqueous solution.

$P + Q \rightleftharpoons R + S$

The following data were obtained.

Substance	Concentration at equilibrium (mol dm^{-3})
P	0.050
Q	0.040
R	0.020
S	0.010

K_c for the reaction is

(A) 0.10.
B 0.33.
C 3.00.
D 10.0.

$K_c = \dfrac{[R][S]}{[P][Q]}$

$= \dfrac{(0.020)(0.010)}{(0.050)(0.040)}$

$= 0.10$

$\therefore$ K_c for the reaction is 0.10.

(Edexcel Advanced GCE, Unit 4, Jan. 2012, 3)

8 4.00 moles of NO(g) and 0.900 mole of $CO_2(g)$ were allowed to react in a closed vessel.

$NO(g) + CO_2(g) \rightleftharpoons NO_2(g) + CO(g)$

At equilibrium, 0.100 mole of $CO_2(g)$ was present. What is the equilibrium constant, K_c, at the temperature of the experiment?

Refer to page T14 for explanation.

A 0.200
B 0.500
(C) 2.00
D 5.00

9 Consider the oxidation of sulphur dioxide to produce sulphur trioxide:

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

At 900 K, the equilibrium constant for the reaction is 13.0 $dm^3\ mol^{-1}$. The equilibrium concentrations are $[O_2(g)] = 0.416\ mol\ dm^{-3}$ and $[SO_3(g)] = 0.840\ mol\ dm^{-3}$. What is the equilibrium concentration of $[SO_2(g)]$?

Refer to page T14 for explanation.

A 0.155 mol dm^{-3}
(B) 0.361 mol dm^{-3}
C 6.44 mol dm^{-3}
D 7.66 mol dm^{-3}

10 At 182 °C, $SbCl_5(g)$ decomposes according to the following equation:

$SbCl_5(g) \rightleftharpoons SbCl_3(g) + Cl_2(g)$

0.300 mole of $SbCl_5(g)$ is placed in a 15.0 dm^3 closed container at 182 °C. 29.4% of $SbCl_5(g)$ decompose when equilibrium is attained.

What is the equilibrium constant, K_c, for this decomposition reaction at 182 °C?

Refer to page T15 for explanation.

A 4.06×10^{-4} mol dm^{-3}
B 1.54×10^{-3} mol dm^{-3}
(C) 2.45×10^{-3} mol dm^{-3}
D 5.88×10^{-2} mol dm^{-3}

11 At 250 °C, the equilibrium constant, K_c, for the following reaction is 25.0 $dm^3\ mol^{-1}$?

$PCl_3(g) + Cl_2(g) \rightleftharpoons PCl_5(g)$

A 5.00 dm^3 sealed container, which is maintained at 250 °C, initially contains 1.00 mole of $PCl_3(g)$ and 1.00 mole of $Cl_2(g)$. What is the concentration of $PCl_5(g)$ when the system attains equilibrium at 250 °C?

Refer to page T15 for explanation.

A 0.833 mol dm^{-3}
B 0.480 mol dm^{-3}
C 0.156 mol dm^{-3}
(D) 0.128 mol dm^{-3}

12 At a certain temperature, the equilibrium constant, K_c, for the following reaction is 1.60×10^{-5} mol dm^{-3}.

$2NOCl(g) \rightleftharpoons 2NO(g) + Cl_2(g)$

A closed flask of volume 1.00 dm^3 initially contains 0.200 mole of NOCl(g), 0.100 mole of NO(g) and 0.100 mole of $Cl_2(g)$.

Which of the following combinations is correct as the system proceeds towards equilibrium?

Refer to page T16 for explanation.

	Reaction quotient (Q_c)	Direction of net reaction
A	$Q_c > K_c$	net forward reaction
B	$Q_c < K_c$	net forward reaction
(C)	$Q_c > K_c$	net backward reaction
D	$Q_c < K_c$	net backward reaction

13 At a certain temperature, it is known that for the reversible reaction:

$A_2(g) + B_2(g) \rightleftharpoons 2AB(g), \quad K_1 = 4.8$

It follows that, for the reaction

$\frac{1}{2}A_2(g) + \frac{1}{2}B_2(g) \rightleftharpoons AB(g), \quad K_2 = X$

X will be

$K_1 = 4.8 = \frac{[AB(g)]^2}{[A_2(g)][B_2(g)]}$

$K_2 = X = \frac{[AB(g)]}{[A_2(g)]^{\frac{1}{2}}[B_2(g)]^{\frac{1}{2}}}$

$X = \sqrt{K_1} = \sqrt{4.8}$

A $\frac{1}{4.8}$.
B 2.4.
C $\frac{1}{2.4}$.
(D) $\sqrt{4.8}$.

14 At a certain temperature, the equilibrium constant, K_c, for the following reaction is 64.0 $dm^6\ mol^{-2}$.

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

A closed container of volume 2.00 dm^3 is filled with 0.280 mole of $N_2(g)$, 0.600 mole of $H_2(g)$ and 0.540 mole of $NH_3(g)$.

Which of the following statements is / are correct as the system proceeds towards equilibrium?

(1) $Q_c < K_c$
(2) Concentration of $H_2(g)$ increases.
(3) Concentration of $NH_3(g)$ decreases.

Refer to page T16 for explanation.

(A) (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

Part III Structured questions

15 A source of hydrogen for the Haber process is 'syngas', which is produced by a reaction of methane and steam at 1 000 °C.

$$CH_4(g) + H_2O(g) \overset{\text{catalyst}}{\rightleftharpoons} CO(g) + 3H_2(g)$$

An equilibrium mixture was found to have the following composition:

$[CH_4(g)] = 2.97 \text{ mol dm}^{-3}$

$[H_2O(g)] = 7.94 \text{ mol dm}^{-3}$

$[CO(g)] = 5.45 \text{ mol dm}^{-3}$

$[H_2(g)] = 2.10 \text{ mol dm}^{-3}$

Calculate the equilibrium constant, K_c, for the reaction.

$K_c = \dfrac{[CO(g)][H_2(g)]^3}{[CH_4(g)][H_2O(g)]}$

$= \dfrac{(5.45 \text{ mol dm}^{-3})(2.10 \text{ mol dm}^{-3})^3}{(2.97 \text{ mol dm}^{-3})(7.94 \text{ mol dm}^{-3})}$

$= 2.14 \text{ mol}^2 \text{ dm}^{-6}$

16 A chemist is investigating the system in which nitrogen monoxide is converted to nitrogen and oxygen.

$$2NO(g) \rightleftharpoons N_2(g) + O_2(g)$$

Initially, she places 1.60 moles of NO(g), 1.60 moles of $N_2(g)$ and 0.600 mole of $O_2(g)$ in a closed vessel of volume 1.00 dm^3. At equilibrium, she finds that the concentration of NO(g) is 1.40 mol dm^{-3}. Calculate the equilibrium constant, K_c, for the reaction.

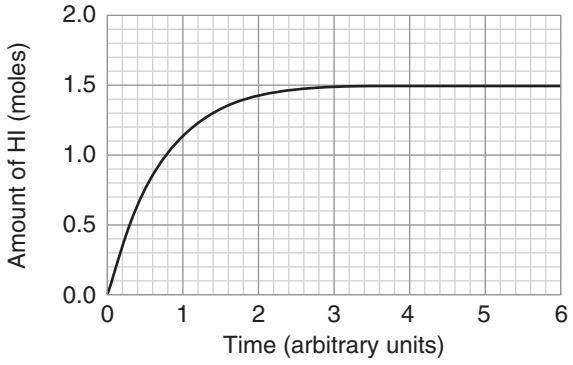
0.607; refer to page T16.

17 Gaseous hydrogen and gaseous iodine react together to form hydrogen iodide.

Refer to page T17.

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

In an experiment, 1.00 mole of $H_2(g)$ and 1.00 mole of $I_2(g)$ were allowed to react inside a closed container of volume 1.00 dm^3. The graph below shows how the amount of HI(g) varies with time in the container.



a) Sketch a similar graph to show how the number of moles of hydrogen varies with time.

b) Write an expression for K_c for the reaction.

c) Calculate the value of K_c for the reaction.

36.0

18 The following dynamic equilibrium was established at temperature T in a closed container.

$$P(g) + 2Q(g) \rightleftharpoons 2R(g)$$

The value of K_c for the reaction was 68.0 dm^3 mol^{-1} when the equilibrium mixture contained 3.82 moles of P and 5.42 moles of R.

Refer to page T17.

a) Give the meaning of the term 'dynamic equilibrium'.

b) Write an expression for K_c for this reaction.

c) The volume of the container was 10.0 dm^3.

Calculate the concentration, in mol dm^{-3}, of Q in the equilibrium mixture.

0.103 mol dm^{-3}

d) Deduce the value of the equilibrium constant, at temperature T, for the reaction below:

0.0147 mol dm^{-3}

$$2R(g) \rightleftharpoons P(g) + 2Q(g)$$

(AQA Advanced GCE, Unit 4, Jun. 2011, 3(a)–(c), (h))

19 2.50 moles of NOCl(g) were placed in a 2.00 dm^3 closed container kept at a constant temperature. After equilibrium was attained, it was found that 28.0% of the NOCl(g) had dissociated.

$$2NOCl(g) \rightleftharpoons 2NO(g) + Cl_2(g)$$

Refer to page T18.

a) How many moles of NOCl(g) remained at equilibrium?

1.80 mol

b) How many moles of NO(g) and $Cl_2(g)$ had been formed at equilibrium?

0.700 mol; 0.350 mol

c) Calculate the equilibrium constant, K_c, for the reaction.

0.0265 mol dm^{-3}

20 The ester $CH_3CH_2COOCH_3$ can be formed from the reaction between propanoic acid and methanol with an acid catalyst.

$$CH_3CH_2COOH + CH_3OH \rightleftharpoons CH_3CH_2COOCH_3 + H_2O$$

a) Name the ester.
Methyl propanoate

b) Complete the table below to show the amount of each substance present at equilibrium. Use your values to calculate the equilibrium constant, K_c, for the reaction. 5.97; refer to page T18.

	Initial amount (mol)	Equilibrium amount (mol)
CH_3CH_2COOH	0.52	0.31
CH_3OH	0.37	0.16
$CH_3CH_2COOCH_3$	0	0.21
H_2O	1.2	1.41

(Edexcel Advanced GCE, Unit 4, Jun. 2012, 17(a)(i), (b))

21 Methanol can be produced by using a reversible reaction between carbon monoxide and hydrogen.

$$2H_2(g) + CO(g) \rightleftharpoons CH_3OH(g)$$

When 2.00 moles of hydrogen and 1.00 mole of carbon monoxide are mixed and heated to a high temperature in a container of volume 1.50 dm^3, the equilibrium yield of methanol is 0.800 mole.

Calculate a value for the equilibrium constant, K_c, for this reaction at this temperature and give its units.
56.3 $dm^6\ mol^{-2}$; refer to page T19.
(AQA Advanced GCE, Unit 5, Jun. 2008, 9(b))

22 Nitrogen and hydrogen react to form ammonia in the presence of a catalyst.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$
$$K_c = 626\ dm^6\ mol^{-2} \text{ at } 200\ °C$$

In an equilibrium mixture, the concentrations of $H_2(g)$ and $NH_3(g)$ are 0.500 mol dm^{-3} and 0.460 mol dm^{-3} respectively. What is the equilibrium concentration of $N_2(g)$?
2.70×10^{-3} mol dm^{-3}; refer to page T19.

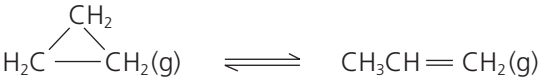
23 For the reaction

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$$

the equilibrium constant at 250 °C is 0.0420 mol dm^{-3}.

A sample of $PCl_5(g)$ was heated to 250 °C. At equilibrium, the concentration of $PCl_5(g)$ was 0.0950 mol dm^{-3}. Calculate the equilibrium concentration of $PCl_3(g)$.
0.0632 mol dm^{-3}; refer to page T19.

24 Cyclopropane gas will undergo isomerization to form propene gas. At a certain temperature, the value of equilibrium constant, K_c, for the isomerization is 5.60.



2.50 moles of cyclopropane are placed in a sealed 0.500 dm^3 container and allowed to come to equilibrium with its isomer, propene.

Determine the concentrations of each gas at equilibrium.
0.760 mol dm^{-3}; 4.24 mol dm^{-3}; refer to page T20.

25 Suppose hydrogen fluoride gas is synthesized by combining 3.00 moles of hydrogen gas and 6.00 moles of fluorine gas in a 3.00 dm^3 flask. The equilibrium constant for the synthesis reaction at this temperature is 1.15×10^2.

What is the concentration of each gas at equilibrium? 1.936 mol dm^{-3}; 0.0320 mol dm^{-3}; 1.032 mol dm^{-3}; refer to page T20.

26 At 1 000 K, the equilibrium constant, K_c, for the following reaction is 2.80×10^2 $dm^3\ mol^{-1}$.

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$$

A mixture of 0.150 mole of $SO_2(g)$, 0.100 mole of $O_2(g)$ and 0.850 mole of $SO_3(g)$ was introduced into an evacuated flask of 1.00 dm^3 kept at 1 000 K.
Refer to page T21.

a) For this system under the initial conditions, calculate its reaction quotient.
321 $dm^3\ mol^{-1}$

b) Predict and explain, under the initial conditions, whether the forward reaction rate or the backward reaction rate is greater.

27 Methanol can be produced by the following reaction:

$$2H_2(g) + CO(g) \rightleftharpoons CH_3OH(g)$$
$$K_c = 10.5\ dm^6\ mol^{-2} \text{ at } 500\ K$$

In a mixture, the concentrations of $H_2(g)$, $CO_2(g)$ and $CH_3OH(g)$ are 0.250 mol dm^{-3}, 0.250 mol dm^{-3} and 0.0400 mol dm^{-3} respectively.

a) Is the system at equilibrium?

b) If not, predict and explain whether the forward reaction rate or the backward reaction rate is greater.

28 The equilibrium constant, K_c, is 0.0900 at 25 °C for the chemical system represented by

$$H_2O(g) + Cl_2O(g) \rightleftharpoons 2HOCl(g)$$

A 3.00 dm^3 flask initially contains 0.500 mole of $H_2O(g)$, 0.100 mole of $Cl_2O(g)$ and 0.250 mole of HOCl(g).
Refer to page T21.

a) i) For this system under the initial conditions, calculate its reaction quotient.
1.25

ii) Predict and explain the direction in which the net reaction will proceed to attain equilibrium.

b) Calculate the concentration of $H_2O(g)$ when the system attains equilibrium at 25 °C.
0.192 mol dm^{-3}

29 Ethanoic acid and ethanol react according to the equation:

$$CH_3COOH(l) + C_2H_5OH(l) \rightleftharpoons CH_3COOC_2H_5(l) + H_2O(l)$$

An experiment was carried out to determine the equilibrium constant, K_c, for the reaction.

Stage 1 0.500 mole of ethanoic acid and 0.500 mole of ethanol were mixed in a pear-shaped flask. 1.00 cm^3 of this mixture was withdrawn and titrated against 0.250 mol dm^{-3} sodium hydroxide solution. 34.0 cm^3 of the alkali were required.

Stage 2 A few drops of concentrated sulphuric acid were added to the mixture in the pear-shaped flask. 1.00 cm^3 of the resulting mixture was withdrawn and titrated against 0.250 mol dm^{-3} sodium hydroxide solution. 34.1 cm^3 of the alkali were required.

Stage 3 The mixture in the pear-shaped flask was heated under reflux for an hour.

Stage 4 The mixture was cooled rapidly in an ice bath.

Stage 5 1.00 cm^3 of the mixture was withdrawn and immediately titrated against 0.250 mol dm^{-3} sodium hydroxide solution. 11.4 cm^3 of the alkali were required.

a) What was the purpose of adding concentrated sulphuric acid in *Stage 2*?
As a catalyst

b) Why was the mixture cooled rapidly in an ice bath in *Stage 4*? To quench the reaction between ethanoic acid and ethanol.

c) Write an expression for K_c for the esterification reaction and calculate its value.
4.01; refer to page T22

d) Suggest what you could do after *Stage 5* to confirm that equilibrium had been attained.
Withdraw 1.00 cm^3 samples from the reaction mixture at different times and titrate against 0.250 mol dm^{-3} sodium hydroxide solution.
The same result would be obtained if equilibrium had been attained.

27 a) $Q_c = \dfrac{[CH_3OH(g)]}{[H_2(g)]^2[CO(g)]}$

$= \dfrac{0.0400\ mol\ dm^{-3}}{(0.250\ mol\ dm^{-3})^2(0.250\ mol\ dm^{-3})}$

$= 2.56\ dm^6\ mol^{-2}$

$Q_c < K_c$, thus the system is not at equilibrium.

b) As $Q_c < K_c$, $[CH_3OH(g)]$ must increase while $[H_2(g)]$ and $[CO(g)]$ must decrease until $Q_c = K_c$.
A net forward reaction occurs, i.e. the forward reaction rate is greater than the backward reaction rate.